



Virtual Reality

Toshiba developed a Helmet in 2006 which gave the user a 360 degree VR environment for gaming. Virtual Reality works on the principle of "Immersion" where 3D images are made to appear life-sized from the perspective of the user along with audio and haptic feedback. Based on the user's motions, particularly his head and eye movements, the images correspondingly adjust on the user's display to reflect the change in perspective. The computer continuously generates an environment for the user and responds to the users every action.

Interfacing with

Amazing interfaces change the way we use technology. Here's a look at the exciting new ways in which we will be interfacing with technology in the future

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Achal's day began with him checking in on his digital identity. Games like Second Life, had evolved into an entire Digital Universe. He was hooked on to his online avatar, which evolved by itself – the latest fad in the gaming scene. All he had to do was wear motion sensors which tracked his movements and actions, while his online self reflected them in the game. In a sense, the concept of parallel universes was not the stuff of science fiction any more. In fact, in the year 2030, there was no such thing as science fiction, there were only anachronisms - the question is never "if" or "how" a technology is possible but rather when?

A new school of thought

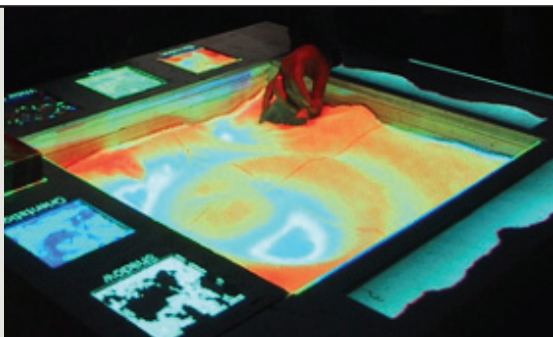
As the clock struck 10 in the morning, Achal walked in to his classroom, or rather, his classroom came to him. The entire class was a single instance of the same virtual reality environment shared by him and his colleagues around the world. His teacher Mr. Morondo was talking about the second world war as he started changing the scenario from a classroom to a battlefield. Percy, Achal's friend, was

interested in primitive weapons and through his hand gestures Percy pointed to various tangible objects in the battlefield and then dragged them to a Wall called the G-Speak.

G-Speak was how everybody computed in the future. Simply put, it was a surface which acted as a base to project a 3D image for the user to manipulate information. He made hand gestures, much like directing an orchestra, to align weapons according to their destructive power in a 3D pyramid, where the bottom layer was constructed of cannonballs made of gunpowder, and at the very top was an atomic bomb. The pyramid was just one among many visualization diagram metaphors that served as a real time representation of data.

At the end of the week, there would of course, be a test (an antique tradition that withstood the test of time). At the day of the test, everybody had to wear devices on their heads that were excruciatingly efficient in what they did – recognising the signals of the brain. An intention to cheat was instantly detected at its inception. His friend Abhishek though, was trying to find some way to work around this little niggle.

Later that day as Achal started walking down the streets. High rise buildings with huge holographic projections rising to the sky from the terrace tops, were telling him to swipe his thumb on his



SandScape

SandScape is a tangible interface developed by the MIT Media Lab, for designing and understanding landscapes through a variety of computational simulations using sand. Users view these simulations as they are projected on the surface of a sand model that represents the terrain. The users can choose from a variety of different simulations that highlight either the height, slope, contours, shadows, drainage or aspect of the landscape model. The users can alter the form of the landscape model by manipulating sand while seeing the resultant effects of computational analysis generated and projected on the surface of sand in real-time.